

Name _____ Teacher _____



GOSFORD HIGH SCHOOL

2012

HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 2 MATHEMATICS – EXTENSION 1

Duration- 90 minutes plus 5 minutes reading time

Question 1 (Multiple choice) 6 questions worth 1 mark each. (Answer this section on the test paper . Circle the correct answer)	/6
Questions 2 to 7 (free response questions) These must be answered on your own paper . Start a new page for each question. Each question must be stapled separately Each question worth 9 marks .	
Question 2 Preliminary topics	/9
Question 3 Iterative methods/polynomials	/9
Question 4 Mathematical induction	/9
Question 5 Parametric representation	/9
Question 6 Methods of integration	/9
Question 7 Circle Geometry	/9
TOTAL	/60

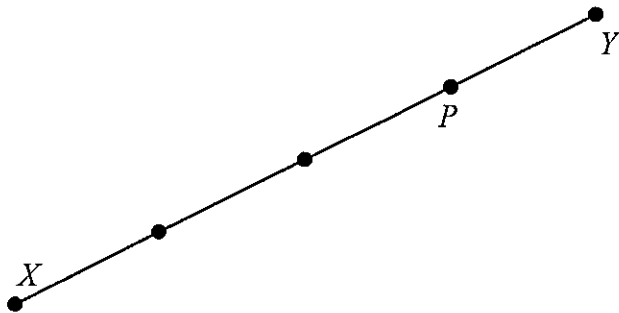
QUESTION 1

6 marks

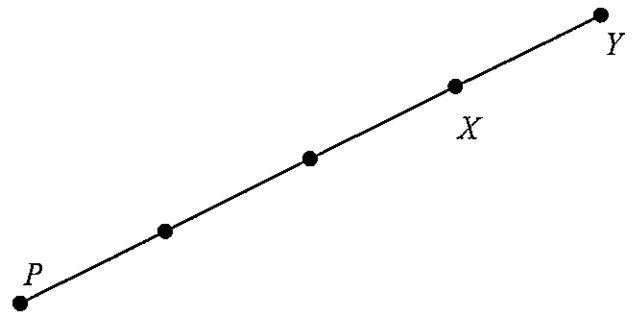
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- i) Which one of the following diagrams best represents a point P dividing an interval XY externally in the ratio $3 : 1$

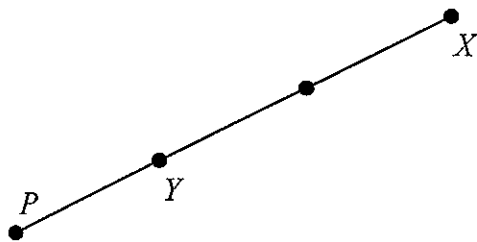
A)



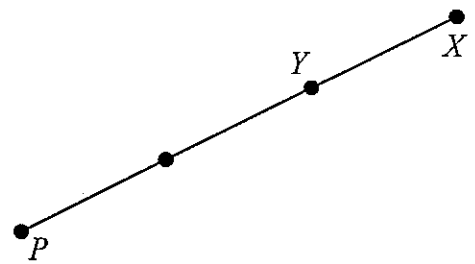
B)



C)



D)



- ii) The curve $y = f(x)$ is defined for all values of x .
Given that $a < b < c$ and K is a constant, which one of the following is NOT necessarily always true.

A) $\int_b^a f(x)dx = -\int_a^b f(x)dx$

B) $\int_c^a K f(x)dx = K \int_c^a f(x)dx$

C) $\int_c^a f(x)dx = \int_b^a f(x)dx + \int_c^b f(x)dx$

D) $\int_{-a}^a f(x)dx = 2 \times \int_0^a f(x)dx$

- iii) If $y = f(x)$ is defined at $x = a$. Which one of the following gives the necessary condition(s) for the curve $y = f(x)$ to be continuous at $x = a$.
- A) $f'(a) = 0$
- B) $f(a) = f'(a)$
- C) $\lim_{x \rightarrow -a} f(x) = f(a) = \lim_{x \rightarrow a} f(x)$
- D) $\lim_{x \rightarrow a} f(x) = f'(a)$
- iv) A parabola has parametric equations $x = 2p$, $y = -2 - 2p^2$. This parabola has :
- A) Vertex $(0, -2)$ and Focus $(0, -2\frac{1}{2})$
- B) Vertex $(0, 2)$ and Focus $(0, -1\frac{1}{2})$
- C) Vertex $(0, -2)$ and Focus $(0, -1\frac{1}{2})$
- D) Vertex $(0, 2)$ and Focus $(0, -2\frac{1}{2})$
- v) The hyperbola $y = \frac{2x-3}{x+6}$ has a :
- A) vertical asymptote with equation $x = -6$ and a horizontal asymptote with equation $y = 1\frac{1}{2}$
- B) vertical asymptote with equation $x = -6$ and a horizontal asymptote with equation $y = 2$
- C) vertical asymptote with equation $x = -6$ and a horizontal asymptote with equation $y = 0$
- D) vertical asymptote with equation $x = -6$ and a horizontal asymptote with equation $y = -\frac{1}{2}$
- vi) A student uses the substitution $y = \sqrt{x}$ to find $\int \frac{dx}{\sqrt{x(1-x)}}$.

Which one of the following represents a equivalent definite integral.

- A) $2 \int \frac{dy}{\sqrt{1-y^2}}$ B) $\frac{1}{2} \int \frac{dy}{\sqrt{1-y^2}}$
- C) $2 \int \frac{y dy}{\sqrt{1-y^2}}$ D) $\frac{1}{2} \int \frac{y dy}{\sqrt{1-y^2}}$

QUESTION 2**9 marks***(start a new page)*

- a) The point $P(-1, -3)$ divides the interval joining $A(5, 9)$ and $B(-3, -7)$ internally in the ratio $k : 1$. Find k (3)
- b) Solve $\frac{4x+3}{2x-1} < 1$ (3)
- c) Find the acute angle between the two curves $y = x^3$ and $y = (2x-1)^2$ at their point of intersection $(1, 1)$. Give your answer correct to the nearest degree. (3)

QUESTION 3**9 marks***(start a new page)*

- a) The equation $2x^3 - 9x^2 + 12x - 4 \cdot 01 = 0$ has a root near $x = 2$
- (i) If $x = 2$ is chosen as the initial approximation, explain why Newton's Method for finding an improved approximation fails. (2)
- (ii) Use Newton's Method once and $x = 1 \cdot 5$ as an initial approximation to the root of $2x^3 - 9x^2 + 12x - 4 \cdot 01 = 0$ to find an improved approximation to the root. (write your answer correct to 2 d.p.) (2)
- b)
- (i) Show that the quadratic polynomial $3x^2 + 2x + 1$ is positive definite. (1)
- (ii) If $P(x) = x^3 + x^2 + x - 8$, use the sign derivative of $P(x)$ to explain why the equation $P(x) = 0$ has only one root for all real values of x . (1)
- (iii) Given that this root lies between $x = 1 \cdot 5$ and $x = 2$, use the Halving the Interval Method for approximating a root once to show that the root lies more accurately between $x = 1 \cdot 5$ and $x = 1 \cdot 75$ (2)
- (iv) Is the root closer to $x = 1 \cdot 5$ or $x = 1 \cdot 75$. Give reasons for your answer. (1)

QUESTION 4**9 marks***(start a new page)*

- a) Show, by process of mathematical induction, that $3^{4n} - 1$ is a multiple of 80 for all positive integral values of n (3)

- b) Use mathematical induction to prove that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \times 2^n \text{ for all positive integral values of } n \quad (4)$$

- c) A student is asked to use Mathematical Induction to prove that $5^n > 3^n + 4^n$ for integers $n \geq 3$.

His incomplete solution to the proof is given below.

Complete Step 3 of the proof on your own paper (2)

Step 1 Prove true for $n = 1$

$$\text{L.H.S.} = 5^n$$

$$= 5^3 \text{ when } n = 3$$

$$= 125$$

$$\text{R.H.S.} = 3^n + 4^n$$

$$= 3^3 + 4^3$$

$$= 91$$

$$\therefore \text{L.H.S.} > \text{R.H.S. for } n = 3$$

$$\therefore 5^n > 3^n + 4^n \text{ for } n = 3$$

Step 2 Assume true for $n = k$

i.e. assume $5^k > 3^k + 4^k$

Step 3 Prove true for $n = k + 1$, if true for $n = k$

i.e. prove $5^{k+1} > 3^{k+1} + 4^{k+1}$

.....

.....

.....

.....

Step 4 Since true for $n = 1$

QUESTION 5

9 marks

(start a new page)

- a) The normal at the point $P(4t, 2t^2)$ on the parabola $x^2 = 8y$ intersects the y axis at Q .
- (i) Show that the gradient of the normal at P is $-\frac{1}{t}$ (1)
- (ii) Show that the equation of the normal at P is $x + ty = 4t + 2t^3$ (1)
- (iii) Find, in terms of t , the coordinates of Q . (1)
- (iv) State the **parametric equations** of the locus of the midpoint of PQ . (1)
- b) The points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$.
The chord PQ , when produced, always passes through the point $A(2, 0)$.
- (i) Noting that $p \neq q$, show that $p + q = pq$. (2)
- (ii) Find the co-ordinates of M , the midpoint of PQ (1)
- (iii) Find the **cartesian equation** of the locus of M (2)

QUESTION 6

9 marks

(start a new page)

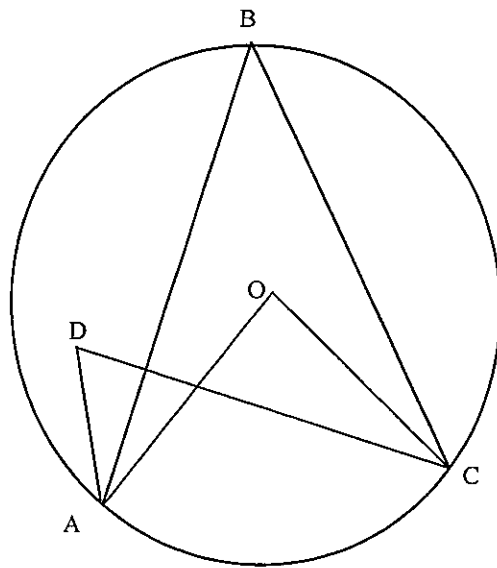
- a) Evaluate the definite integral $\int \frac{t \, dt}{\sqrt{4-t}}$ using the substitution $t = 4 - w^2$ (3)
- b) Find $\int \frac{x \, dx}{\sqrt{1-9x^2}}$ using the substitution $u = \sqrt{1-9x^2}$ (3)
- c) Use the substitution $u = x+1$ to evaluate $\int_0^3 \frac{x-2}{\sqrt{x+1}} \, dx$ (3)

QUESTION 7

9 marks

(start a new page)

a)

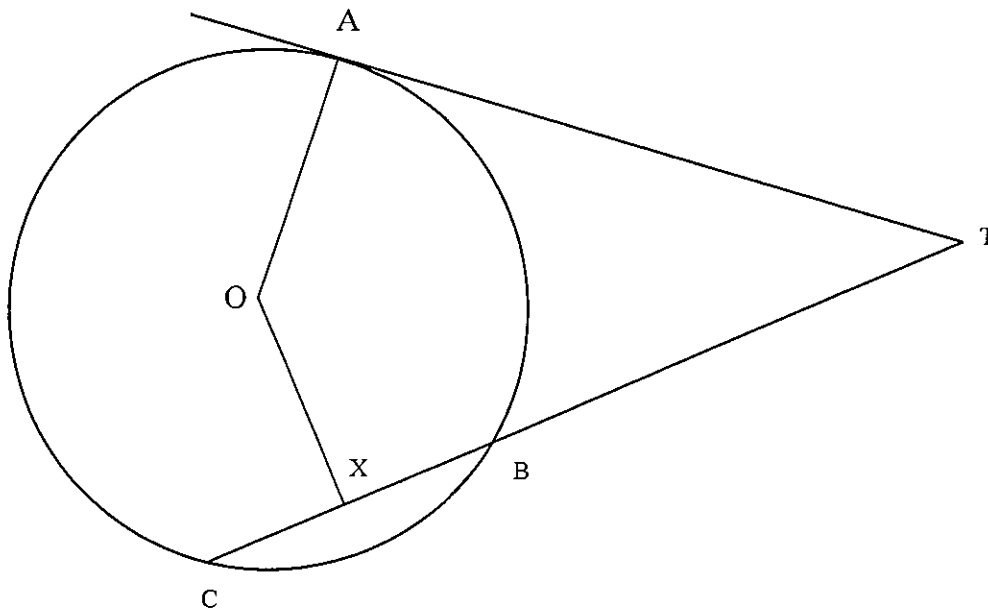


Points A , B and C lie on the circumference of a circle with centre O .
The point D lies inside the circle. $\angle ABC = 17^\circ$ and $\angle ADC = 34^\circ$

(i) Find $\angle AOC$ giving reasons. (1)

(ii) State why $ADOC$ is a cyclic quadrilateral. (1)

b)

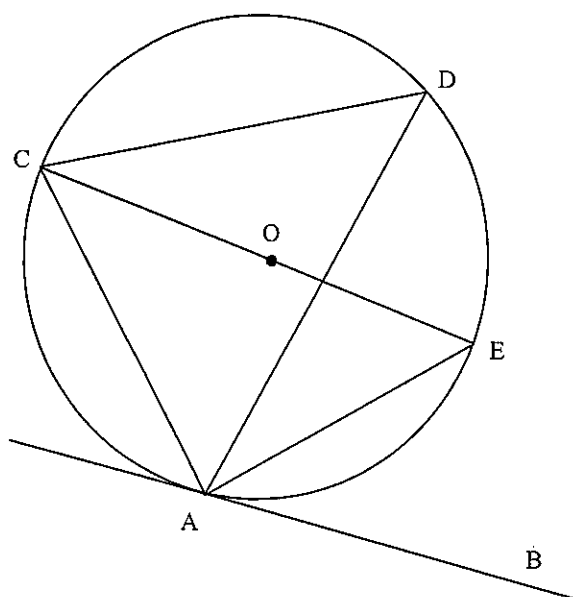


A , B and C are three points on a circle with centre O .

The tangent at A meets the chord CB produced at T . X is the midpoint of CB .

Prove that $AOXT$ is a cyclic quadrilateral. (3)

c)



AB is a tangent and CE is a diameter to a circle with centre O .

$\angle BAE$ is 48° and D lies on the circumference as show in the diagram.

- (i) Find the size of $\angle ACE$, giving reasons. (1)
- (ii) Find the size of $\angle ADC$, justifying your answer with reasoning. (3)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONSQuestion 1

- (i) C (ii) D (iii) C
 (iv) A (v) B (vi) A

Question 2

- a) A(5, 9) B(-3, -7)

$$k : 1$$

$$\therefore \frac{-3k+5}{k+1} = -1$$

$$-3k+5 = -k-1$$

$$6 = 2k$$

$$\therefore k = 3$$

b) Critical values occur when

$$2x-1=0$$

$$x = \frac{1}{2}$$

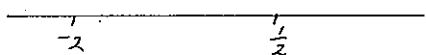
and when

$$\frac{4x+3}{2x-1} = 1$$

$$4x+3 = 2x-1$$

$$2x = -4$$

$$x = -2$$

FALSE When $x < -2$ (say $x = -3$)TRUE When $-2 < x < \frac{1}{2}$ (say $x = 0$)FALSE When $x > \frac{1}{2}$ (say $x = 1$) $\therefore$ Solution is $-2 < x < \frac{1}{2}$

c)

$$y = x^3$$

$$\frac{dy}{dx} = 3x^2$$

$$m_1 = 3 \text{ when } x = 1$$

$$y = (2x-1)^2$$

$$\frac{dy}{dx} = 2(2x-1) \cdot 2$$

$$m_2 = 4$$

$$\text{when } x = 1$$

Let θ be the required acute $\angle$.

$$\therefore \tan \theta = \left| \frac{m_2 - m_1}{1 + m_2 m_1} \right|$$

$$\tan \theta = \left| \frac{4 - 3}{1 + (4)(3)} \right|$$

$$\tan \theta = \frac{1}{13}$$

$$\theta = 4^\circ \text{ (to the nearest degree)}$$

Question 3(a) (i) Let $P(x) = 2x^3 - 9x^2 + 12x - 4.01$

$$P'(x) = 6x^2 - 18x + 12$$

$$P'(2) = 24 - 36 + 12$$

$$= 0$$

 $\therefore$ Stationary Pt. exists at $x = 2$

$\therefore$ Tangent at $x = 2$ does not cross the x -axis.
 The x -intercept of the tangent is the improved approximation.
 Thus an improved approximation cannot be found.

(ii) If $x_1 = 1.5$ is the initial approx.

$$P(x_1) = P(1.5)$$

$$= 0.49$$

$$P'(x_1) = P'(1.5)$$

$$= -1.5$$

If x_2 is improved approx. then $x_2 = x_1 - \frac{P(x_1)}{P'(x_1)}$

$$= 1.5 + \frac{0.49}{1.5}$$

$$= 1.83 \text{ (to 2d.p.)}$$

$$\begin{aligned} b) \quad (i) \quad \Delta &= b^2 - 4ac \\ &= 4 - 4(3)(1) \\ &= -8 < 0 \end{aligned}$$

Since $a = 3 > 0$ and $\Delta < 0$ expression is +ve definite

$$(ii) \quad P'(x) = 3x^2 + 2x + 1 > 0$$

$\therefore$ Curve is always increasing
Since $P(x)$ is continuous for all x
the equation $P(x) = 0$ has only one root

$$(iii) \quad P(1.5) = -0.875 < 0 \quad P(2) = 6$$

Let $x_0 = \frac{1.5+2}{2}$ be new approx.

$$x_0 = 1.75$$

$$P(1.75) = 2.171875 > 0$$

$P(1.75)$ and $P(1.5)$ are opposite in sign
 $\therefore$ Root lies between $x = 1.5$ and $x = 1.75$

$$\begin{aligned} (iv) \quad \text{Consider } P\left(\frac{1.5+1.75}{2}\right) &= P(1.625) \\ &= 0.55 > 0 \end{aligned}$$

$\therefore$ Root lies between $x = 1.625$ and $x = 1.5$
 $\therefore$ Root is closer $x = 1.5$ than $x = 1.75$

Question 4

a) Prove true for $n=1$

$$\begin{aligned} 3^{4n} - 1 &= 3^4 - 1 \quad \text{when } n=1 \\ &= 80 \end{aligned}$$

Which is a multiple of 80
 $\therefore$ true for $n=1$

Assume true for $n=k$

$$\begin{aligned} \text{i.e. } 3^{4k} - 1 &= 80M \quad (\text{where } M \text{ is a +ve integer}) \\ 3^{4k} &= 80M + 1 \end{aligned}$$

Prove true for $n=k+1$, if true for $n=k$

$$\begin{aligned} \text{i.e. Prove } 3^{4(k+1)} - 1 &\text{ is a multiple of 80} \\ 3^{4(k+1)} - 1 &= 3^{4k+4} - 1 \\ &= 3^4 \cdot 3^{4k} - 1 \\ &= 81(80M+1) - 1 \quad \text{using assumption} \\ &= 81 \times 80M + 81 - 1 \\ &= 80(81M+1) \end{aligned}$$

Which is a multiple of 80 since
 $(81M+1)$ is a +ve integer

$\therefore$ true for $n=k+1$, if true for $n=k$

Since true for $n=1 \therefore$ true for $n=1+1=2$

Since true for $n=2 \therefore$ true for $n=2+1=3$

and so on.

$\therefore$ true for all given 'n'

Question 4 (b) Prove true for $n=1$

$$L.H.S. = n \times 2^{n-1}$$

$$= 1 \times 2^0 \text{ when } n=1$$

$$= 1$$

$\therefore$ true for $n=1$

$$R.H.S. = 1 + (n-1) \times 2^n$$

$$= 1 + 0 \text{ when } n=1$$

$$= 1$$

$$= L.H.S$$

Assume true for $n=k$

$$i.e. 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1) \times 2^k$$

Prove true for $n=k+1$, if true for $n=k$

$$i.e. \text{ Prove } 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1) \times 2^k = 1 + k \times 2^{k+1}$$

$$L.H.S. = 1 + (k-1) \times 2^k + (k+1) \times 2^k \text{ using assumption}$$

$$= 1 + 2^k (k-1+k+1)$$

$$= 1 + 2^k \times 2k$$

$$= 1 + k \times 2^{k+1}$$

$$= R.H.S.$$

$\therefore$ true for $n=k+1$ if true for $n=k$

Since true for $n=1$ $\therefore$ true for $n=1+1=2$

Since true for $n=2$ $\therefore$ true for $n=2+1=3$

and so on

$\therefore$ true for all give 'n'

Question 4 c)

$$5^{k+1} = 5 \times 5^k$$

$$\therefore 5^{k+1} > 5(3^k + 4^k) \text{ using assumption}$$

$$5^{k+1} > 5 \times 3^k + 5 \times 4^k$$

$$5^{k+1} > 3 \times 3^k + 4 \times 4^k \text{ since } 3 < 5 \text{ and } 4 < 5$$

$$5^{k+1} > 3^{k+1} + 4^{k+1}$$

$\therefore$ true for $n=k+1$ if true for $n=k$

Question 5

$$a) (i) y = \frac{1}{8} x^2$$

$$\frac{dy}{dx} = \frac{1}{4} x$$

$$= \frac{1}{4} \times 4t \text{ at } P$$

$$= t$$

(gradient of tangent at P)

$\therefore$ gradient of normal at P is $-\frac{1}{t}$

$$(ii) \text{ Equation of normal is } y - 2t^2 = -\frac{1}{t}(x - 4t)$$

$$i.e. ty - 2t^3 = -x + 4t$$

$$x + ty = 4t + 2t^3$$

$$(iii) \text{ At } y \text{ axis } x = 0$$

$$\therefore ty = 4t + 2t^3$$

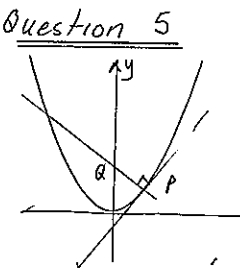
$$y = 4 + 2t^2$$

$$\therefore Q(0, 4 + 2t^2)$$

$$(iv) \text{ Midpoint of } PQ \text{ is } \left(\frac{4t}{2}, \frac{2t^2 + 4 + 2t^2}{2} \right)$$

$$= (2t, 2t^2 + 2)$$

$\therefore$ Parametric equations are $x = 2t$, $y = 2t^2 + 2$



$$b) \quad (i) \quad m_{AP} = m_{AQ}$$

$$\frac{p^2}{2p-2} = \frac{q^2}{2q-2}$$

$$2p^2q - 2p^2 = 2q^2p - 2q^2$$

$$2p^2q - 2q^2p = 2p^2 - 2q^2$$

$$2pq(p-q) = 2(p-q)(p+q)$$

$$pq = p+q \quad \text{since } p \neq q$$

$$(ii) \quad M \text{ is } \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$$

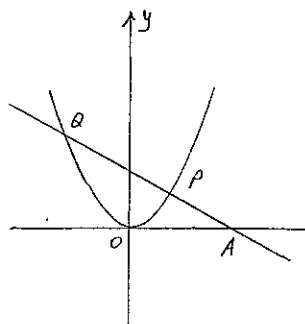
$$= \left(p+q, \frac{p^2+q^2}{2} \right)$$

(iii) Parametric equations are

$$x = p+q \quad \text{and} \quad y = (p+q)^2 - 2pq$$

$$\therefore y = x^2 - 2(p+q) \quad \text{using (i)}$$

$$y = x^2 - 2x \quad \text{is the cartesian equation}$$



Question 6

$$a) \quad t = 4 - w^2 \rightarrow w^2 = 4 - t$$

$$\frac{dt}{dw} = -2w$$

$$\therefore \int \frac{t dt}{\sqrt{4-t}} = \int \frac{4-w^2}{\sqrt{w^2}} \times -2w dw$$

$$= 2 \int w^2 - 4 dw$$

$$= 2 \left[\frac{w^3}{3} - 4w \right] + C$$

$$= \frac{2}{3} \sqrt{(4-t)^3} - 8\sqrt{4-t} + C$$

$$b) \quad u = (1-9x^2)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2} (1-9x^2)^{-\frac{1}{2}} \times -18x$$

$$= \frac{-9x}{\sqrt{1-9x^2}}$$

$$\frac{dx}{du} = \frac{u}{-9x}$$

$$\therefore \int \frac{x dx}{\sqrt{1-9x^2}} = \int \frac{x}{u} \times \frac{u}{-9x} du$$

$$= -\frac{1}{9} \int 1 du$$

$$= -\frac{1}{9} u + C$$

$$= -\frac{1}{9} \sqrt{1-9x^2} + C$$

Question 6

c) $u = x+1$ when $x=3$, $u=4$

$$\frac{du}{dx} = 1 \rightarrow \frac{dx}{du} = 1 \quad x=0, u=1$$

$$\begin{aligned}
 \therefore \int_0^3 \frac{x-2}{\sqrt{x+1}} dx &= \int_1^4 \frac{u-3}{\sqrt{u}} \times 1 du \\
 &= \int_1^4 \frac{u}{u^{\frac{1}{2}}} - \frac{3}{u^{\frac{1}{2}}} du \\
 &= \int_1^4 u^{\frac{1}{2}} - 3u^{-\frac{1}{2}} \\
 &= \left[\frac{2u^{\frac{3}{2}}}{3} - 6u^{\frac{1}{2}} \right]_1^4 \\
 &= \left[\frac{2}{3} \times 8 - 6 \times 2 - \left(\frac{2}{3} - 6 \right) \right] \\
 &= \frac{14}{3} - 6 \\
 &= -\frac{4}{3}
 \end{aligned}$$

Question 7

a)

(i) $\angle AOC = 2 \times \angle ABC$ ($\angle$ at the centre is twice the angle at the circumference, standing on the same arc)
 $= 2 \times 17^\circ$
 $= 34^\circ$

(ii) $\angle AOC = \angle ADC$
 $= 34^\circ$

Since the interval AC subtends two equal $\angle$'s at O and D then the end points of the interval, A & C and the vertices of the angles, O & D are concyclic i.e. ADOC is a cyclic quadrilateral

Question 7

b) $\angle AOT = 90^\circ$ (tangent is perpendicular to the radius drawn at the point of contact)

$\angle AXB = 90^\circ$ (interval drawn from the centre of a circle to the midpoint of a chord is perpendicular to the chord)

$\therefore \angle AOT + \angle AXB = 180^\circ$

Since the opposite angles of the quadrilateral AOTX are supplement, AOTX must be a cyclic quadrilateral.

c) (i) $\angle ACE = \angle ABE$ ($\angle$ between a tangent and a chord drawn at the point of contact is equal to the angle in the alternate segment)
 $= 48^\circ$

(ii) $\angle CAE = 90^\circ$ (angle in a semi-circle is a right angle)

$\angle CEA = 180 - (90 + 48)^\circ$ ($\angle$ sum of $\triangle ACE$)
 $= 42^\circ$

$\angle ADC = \angle CEA$ (angles at the circumference standing on the same arc are equal)